

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Data Structures and Algorithms [ECE_4447]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) What are the properties to be satisfied for an algorithm. Discuss each with an example. (4)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Illustrate and explain space complexity using suitable code. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Suggest an algorithm to find the transpose of a Sparse Matrix. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Illustrate and explain the postfix expression evaluation in stacks. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Discuss the implementation of Queues using linked lists. (3)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
C) With an example code, explain the deletion operation at the end for a Singly linked list. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Illustrate and explain the methods for constructing an expression tree. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Mention tree traversal operations and explain Inorder traversal with suitable example. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Prove that the height of AVL tree with 'n' elements is $O(\log n)$ (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) With an illustration, explain the adjacency matrix to represent graphs. (4)
A) Course Outcome: "CO4" / Cognitive Level: "Application"

B) Suggest an algorithm for Binary search considering both Best case and Worst case complexities. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) Discuss the steps of Quick sort algorithm. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) Differentiate between Linear Search and Binary Search mentioning the significance of both. (4)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Discuss the steps involved in merge sort. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

C) Elaborate the steps involved in Breadth First Search (BFS) Algorithm (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4447:Data Structures and Algorithms

#	Course Outcome	Description	Marks
1	1	CO 1	4
2	2	CO 2	6
3	3	CO 3	20
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4447:Data Structures and Algorithms

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	10(20%)
3	Application	8	27(54%)
4	Analysis	4	13(26%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)